Crash Lab — Collisions Transfer Energy

Chapter 5 · Collisions Transfer Energy · Week 3

Name: _____ Date: _____

BIG QUESTION

What happens when objects collide? How are collisions predictable?

OUR MODEL

COLLISION → MOTION CAN CHANGE → ENERGY CAN TRANSFER → ENERGY DOES NOT DISAPPEAR

PART A — SPEED

FAIR TEST: SAME car, SAME target, SAME surface. We change ONLY the ramp height (1 / 2 / 3 books). A higher ramp means a faster car.

PREDICT:

At which ramp height do you think the target will move the farthest? Why?

TEST: roll the SAME car down each ramp height. Measure how far the target moves (cm). 3 trials each.

Condition	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)	Sound (loud / quiet)
1 book					
2 books					
3 books					

COMPARE: At which ramp height did the target move the farthest?

EXPLAIN: How did the SPEED of the car affect the change it caused?

PART B — MASS / WEIGHT (OPTIONAL extension — separate fair test)

FAIR TEST: SAME ramp height, SAME target, SAME surface. We change ONLY how much the car weighs (1 car / 2 cars taped / 3 cars taped). The cars roll at about the same speed.

PREDICT:

Which car do you think will move the target the farthest? Why?

TEST: roll each mass down the SAME ramp height. Measure how far the target moves (cm). 3 trials each.

Condition	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)	Sound (loud / quiet)
1 car					
2 cars					
3 cars					

COMPARE: Which mass moved the target the farthest?

EXPLAIN: How did the WEIGHT of the car affect the change it caused?

PUT IT TOGETHER

Did the energy disappear when the car hit the target? Where did the energy go?